

9.1

(a)

The function $f(x) = 1/\sqrt[5]{x} = x^{-1/5}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test applies.

$$\int_1^\infty x^{-1/5} dx = \lim_{t \rightarrow \infty} \int_1^t x^{-1/5} dx = \lim_{t \rightarrow \infty} \left[\frac{5}{4} x^{4/5} \right]_1^t = \lim_{t \rightarrow \infty} \left(\frac{5}{4} t^{4/5} - \frac{5}{4} \right) = \infty, \text{ so } \sum_{n=1}^\infty 1/\sqrt[5]{n} \text{ diverges.}$$

(b)

The function $f(x) = \frac{1}{(2x+1)^3}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test applies.

$$\int_1^\infty \frac{1}{(2x+1)^3} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{(2x+1)^3} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{4} \frac{1}{(2x+1)^2} \right]_1^t = \lim_{t \rightarrow \infty} \left(-\frac{1}{4(2t+1)^2} + \frac{1}{36} \right) = \frac{1}{36}.$$

Since this improper integral is convergent, the series $\sum_{n=1}^\infty \frac{1}{(2n+1)^3}$ is also convergent by the Integral Test.

(c)

The function $f(x) = \frac{x}{x^2+1}$ is continuous, positive, and decreasing on $[1, \infty)$, so the Integral Test applies.

$$\int_1^\infty \frac{x}{x^2+1} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{x}{x^2+1} dx = \lim_{t \rightarrow \infty} \left[\frac{1}{2} \ln(x^2+1) \right]_1^t = \frac{1}{2} \lim_{t \rightarrow \infty} [\ln(t^2+1) - \ln 2] = \infty. \text{ Since this improper}$$

integral is divergent, the series $\sum_{n=1}^\infty \frac{n}{n^2+1}$ is also divergent by the Integral Test.

(d)

The function $f(x) = x^2 e^{-x^3}$ is continuous, positive, and decreasing ($\star$) on $[1, \infty)$, so the Integral Test applies.

$$\int_1^\infty x^2 e^{-x^3} dx = \lim_{t \rightarrow \infty} \int_1^t x^2 e^{-x^3} dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{3} e^{-x^3} \right]_1^t = -\frac{1}{3} \lim_{t \rightarrow \infty} (e^{-t^3} - e^{-1}) = -\frac{1}{3} \left(0 - \frac{1}{e} \right) = \frac{1}{3e}.$$

Since this improper integral is convergent, the series $\sum_{n=1}^\infty n^2 e^{-n^3}$ is also convergent by the Integral Test.

$$(\star): f'(x) = x^2 e^{-x^3}(-3x^2) + e^{-x^3}(2x) = x e^{-x^3}(-3x^3 + 2) = \frac{x(2-3x^3)}{e^{x^3}} < 0 \text{ for } x > 1$$

9.2

Clearly, if $p < 0$ then the series diverges, since $\lim_{n \rightarrow \infty} \frac{1}{n^p \ln n} = \infty$. If $0 \leq p \leq 1$, then $n^p \ln n \leq n \ln n \Rightarrow$

$\frac{1}{n^p \ln n} \geq \frac{1}{n \ln n}$ and $\sum_{n=2}^\infty \frac{1}{n \ln n}$ diverges (Exercise 11.3.21), so $\sum_{n=2}^\infty \frac{1}{n^p \ln n}$ diverges. If $p > 1$, use the Limit Comparison

Test with $a_n = \frac{1}{n^p \ln n}$ and $b_n = \frac{1}{n^p}$. $\sum_{n=2}^\infty b_n$ converges, and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1}{\ln n} = 0$, so $\sum_{n=2}^\infty \frac{1}{n^p \ln n}$ also converges.

(Or use the Comparison Test, since $n^p \ln n > n^p$ for $n > e$.) In summary, the series converges if and only if $p > 1$.

9.3

Since $\sum a_n$ converges, $\lim_{n \rightarrow \infty} a_n = 0$, so there exists N such that $|a_n - 0| < 1$ for all $n > N \Rightarrow 0 \leq a_n < 1$ for all $n > N \Rightarrow 0 \leq a_n^2 \leq a_n$. Since $\sum a_n$ converges, so does $\sum a_n^2$ by the Comparison Test.

9.4

(a)

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+1}{5^{n+1}} \cdot \frac{5^n}{n} \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{5} \cdot \frac{n+1}{n} \right| = \frac{1}{5} \lim_{n \rightarrow \infty} \frac{1+1/n}{1} = \frac{1}{5}(1) = \frac{1}{5} < 1$, so the series $\sum_{n=1}^{\infty} \frac{n}{5^n}$ is absolutely convergent by the Ratio Test.

(b)

$b_n = \frac{n}{n^2 + 4} > 0$ for $n \geq 1$, $\{b_n\}$ is decreasing for $n \geq 2$, and $\lim_{n \rightarrow \infty} b_n = 0$, so $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2 + 4}$ converges by the

Alternating Series Test. To determine absolute convergence, choose $a_n = \frac{1}{n}$ to get

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1/n}{n/(n^2 + 4)} = \lim_{n \rightarrow \infty} \frac{n^2 + 4}{n^2} = \lim_{n \rightarrow \infty} \frac{1 + 4/n^2}{1} = 1 > 0$, so $\sum_{n=1}^{\infty} \frac{n}{n^2 + 4}$ diverges by the Limit

Comparison Test with the harmonic series. Thus, the series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2 + 4}$ is conditionally convergent.

(c)

$\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$ converges by the Alternating Series Test since $\lim_{n \rightarrow \infty} \frac{1}{\ln n} = 0$ and $\left\{ \frac{1}{\ln n} \right\}$ is decreasing. Now $\ln n < n$, so

$\frac{1}{\ln n} > \frac{1}{n}$, and since $\sum_{n=2}^{\infty} \frac{1}{n}$ is the divergent (partial) harmonic series, $\sum_{n=2}^{\infty} \frac{1}{\ln n}$ diverges by the Comparison Test. Thus,

$\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$ is conditionally convergent.

(d)

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{[2(n+1)]!}{[(n+1)!]^2} \cdot \frac{(n!)^2}{(2n)!} \right| = \lim_{n \rightarrow \infty} \frac{(2n+2)(2n+1)}{(n+1)(n+1)} = \lim_{n \rightarrow \infty} \frac{(2+2/n)(2+1/n)}{(1+1/n)(1+1/n)} = \frac{2 \cdot 2}{1 \cdot 1} = 4 > 1$,

so the series $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2}$ diverges by the Ratio Test.

9.5

(a)

If $a_n = (-1)^n n x^n$, then

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (n+1) x^{n+1}}{(-1)^n n x^n} \right| = \lim_{n \rightarrow \infty} \left| (-1) \frac{n+1}{n} x \right| = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n} \right) |x| \right] = |x|. \text{ By the Ratio Test, the}$$

series $\sum_{n=1}^{\infty} (-1)^n n x^n$ converges when $|x| < 1$, so the radius of convergence $R = 1$. Now we'll check the endpoints, that is,

$x = \pm 1$. Both series $\sum_{n=1}^{\infty} (-1)^n n (\pm 1)^n = \sum_{n=1}^{\infty} (\mp 1)^n n$ diverge by the Test for Divergence since $\lim_{n \rightarrow \infty} |(\mp 1)^n n| = \infty$. Thus,

the interval of convergence is $I = (-1, 1)$.

(b)

If $a_n = \frac{(-1)^n x^n}{n^2}$, then

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{n+1}}{(n+1)^2} \cdot \frac{n^2}{(-1)^n x^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1) x n^2}{(n+1)^2} \right| = \lim_{n \rightarrow \infty} \left[\left(\frac{n}{n+1} \right)^2 |x| \right] = 1^2 \cdot |x| = |x|.$$

When $x = 1$, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ converges by the Alternating Series Test. When $x = -1$, the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges

since it is a p -series with $p = 2 > 1$. Thus, the interval of convergence is $[-1, 1]$.

(c)

$$\text{If } a_n = (-1)^n \frac{x^n}{4^n \ln n}, \text{ then } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{4^{n+1} \ln(n+1)} \cdot \frac{4^n \ln n}{x^n} \right| = \frac{|x|}{4} \lim_{n \rightarrow \infty} \frac{\ln n}{\ln(n+1)} = \frac{|x|}{4} \cdot 1$$

[by l'Hospital's Rule] $= \frac{|x|}{4}$. By the Ratio Test, the series converges when $\frac{|x|}{4} < 1 \Leftrightarrow |x| < 4$, so $R = 4$. When

$x = -4$, $\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{4^n \ln n} = \sum_{n=2}^{\infty} \frac{[(-1)(-4)]^n}{4^n \ln n} = \sum_{n=2}^{\infty} \frac{1}{\ln n}$. Since $\ln n < n$ for $n \geq 2$, $\frac{1}{\ln n} > \frac{1}{n}$ and $\sum_{n=2}^{\infty} \frac{1}{n}$ is the

divergent harmonic series (without the $n = 1$ term), $\sum_{n=2}^{\infty} \frac{1}{\ln n}$ is divergent by the Comparison Test. When $x = 4$,

$$\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{4^n \ln n} = \sum_{n=2}^{\infty} (-1)^n \frac{1}{\ln n}, \text{ which converges by the Alternating Series Test. Thus, } I = (-4, 4].$$

(d)

$$\text{If } a_n = \frac{x^{2n}}{n (\ln n)^2}, \text{ then } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{2n+2}}{(n+1) [\ln(n+1)]^2} \cdot \frac{n (\ln n)^2}{x^{2n}} \right| = |x^2| \lim_{n \rightarrow \infty} \frac{n (\ln n)^2}{(n+1) [\ln(n+1)]^2} = x^2.$$

By the Ratio Test, the series $\sum_{n=2}^{\infty} \frac{x^{2n}}{n (\ln n)^2}$ converges when $x^2 < 1 \Leftrightarrow |x| < 1$, so $R = 1$. When $x = \pm 1$, $x^{2n} = 1$, the

series $\sum_{n=2}^{\infty} \frac{1}{n (\ln n)^2}$ converges by the Integral Test (see Exercise 11.3.22). Thus, the interval of convergence is $I = [-1, 1]$.

9.6

(a)

$f(x) = \frac{3}{x^2 - x - 2} = \frac{3}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1} \Rightarrow 3 = A(x+1) + B(x-2)$. Let $x = 2$ to get $A = 1$ and $x = -1$ to get $B = -1$. Thus

$$\begin{aligned} \frac{3}{x^2 - x - 2} &= \frac{1}{x-2} - \frac{1}{x+1} = \frac{1}{-2} \left(\frac{1}{1 - (x/2)} \right) - \frac{1}{1 - (-x)} = -\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x}{2} \right)^n - \sum_{n=0}^{\infty} (-x)^n \\ &= \sum_{n=0}^{\infty} \left[-\frac{1}{2} \left(\frac{1}{2} \right)^n - 1(-1)^n \right] x^n = \sum_{n=0}^{\infty} \left[(-1)^{n+1} - \frac{1}{2^{n+1}} \right] x^n \end{aligned}$$

We represented f as the sum of two geometric series; the first converges for $x \in (-2, 2)$ and the second converges for $(-1, 1)$.

Thus, the sum converges for $x \in (-1, 1) = I$.

(b)

$f(x) = \frac{x+2}{2x^2 - x - 1} = \frac{x+2}{(2x+1)(x-1)} = \frac{A}{2x+1} + \frac{B}{x-1} \Rightarrow x+2 = A(x-1) + B(2x+1)$. Let $x = 1$ to get $3 = 3B \Rightarrow B = 1$ and $x = -\frac{1}{2}$ to get $\frac{3}{2} = -\frac{3}{2}A \Rightarrow A = -1$. Thus,

$$\begin{aligned} \frac{x+2}{2x^2 - x - 1} &= \frac{-1}{2x+1} + \frac{1}{x-1} = -1 \left(\frac{1}{1 - (-2x)} \right) + 1 \left(\frac{1}{1 - x} \right) = -\sum_{n=0}^{\infty} (-2x)^n + \sum_{n=0}^{\infty} x^n \\ &= -\sum_{n=0}^{\infty} [(-2)^n + 1] x^n \end{aligned}$$

We represented f as the sum of two geometric series; the first converges for $x \in (-\frac{1}{2}, \frac{1}{2})$ and the second converges for $(-1, 1)$. Thus, the sum converges for $x \in (-\frac{1}{2}, \frac{1}{2}) = I$.

9.7

(a)

$$f(x) = \ln(5-x) = - \int \frac{dx}{5-x} = -\frac{1}{5} \int \frac{dx}{1-x/5} = -\frac{1}{5} \int \left[\sum_{n=0}^{\infty} \left(\frac{x}{5} \right)^n \right] dx = C - \frac{1}{5} \sum_{n=0}^{\infty} \frac{x^{n+1}}{5^n(n+1)} = C - \sum_{n=1}^{\infty} \frac{x^n}{n 5^n}$$

Putting $x = 0$, we get $C = \ln 5$. The series converges for $|x/5| < 1 \Leftrightarrow |x| < 5$, so $R = 5$.

(b)

$$f(x) = x^2 \tan^{-1}(x^3) = x^2 \sum_{n=0}^{\infty} (-1)^n \frac{(x^3)^{2n+1}}{2n+1} \quad [\text{by Example 7}] = \sum_{n=0}^{\infty} (-1)^n \frac{x^{6n+3+2}}{2n+1} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{6n+5}}{2n+1} \text{ for}$$

$$|x^3| < 1 \Leftrightarrow |x| < 1, \text{ so } R = 1.$$

(c)

We know that $\frac{1}{1+4x} = \frac{1}{1-(-4x)} = \sum_{n=0}^{\infty} (-4x)^n$. Differentiating, we get

$$\frac{-4}{(1+4x)^2} = \sum_{n=1}^{\infty} (-4)^n n x^{n-1} = \sum_{n=0}^{\infty} (-4)^{n+1} (n+1) x^n, \text{ so}$$

$$f(x) = \frac{x}{(1+4x)^2} = \frac{-x}{4} \cdot \frac{-4}{(1+4x)^2} = \frac{-x}{4} \sum_{n=0}^{\infty} (-4)^{n+1} (n+1) x^n = \sum_{n=0}^{\infty} (-1)^n 4^n (n+1) x^{n+1}$$

$$\text{for } |-4x| < 1 \Leftrightarrow |x| < \frac{1}{4}, \text{ so } R = \frac{1}{4}.$$

(d)

$$f(x) = \frac{x^2-x}{(1-x)^3} = \frac{-x(1-x)}{(1-x)^3} = -\frac{x}{(1-x)^2}$$

Differentiating each side of the eq

$$\frac{1}{1-x} = 1+x+x^2+\dots = \sum_{n=0}^{\infty} x^n$$

we get

$$\frac{1}{(1-x)^2} = 1+2x+3x^2+\dots = \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=0}^{\infty} (n+1) x^n$$

$$\text{So, } f(x) = -x \sum_{n=0}^{\infty} (n+1) x^n = -\sum_{n=1}^{\infty} n x^n$$

with $R=1$

9.8

(a)

n	$f^{(n)}(x)$	$f^{(n)}(-2)$
0	$x - x^3$	6
1	$1 - 3x^2$	-11
2	$-6x$	12
3	-6	-6
4	0	0
5	0	0
$\vdots$	$\vdots$	$\vdots$

$f^{(n)}(x) = 0$ for $n \geq 4$, so f has a finite series expansion about $a = -2$.

$$\begin{aligned}
 f(x) &= x - x^3 = \sum_{n=0}^3 \frac{f^{(n)}(-2)}{n!} (x+2)^n \\
 &= \frac{6}{0!} (x+2)^0 + \frac{-11}{1!} (x+2)^1 + \frac{12}{2!} (x+2)^2 + \frac{-6}{3!} (x+2)^3 \\
 &= 6 - 11(x+2) + 6(x+2)^2 - (x+2)^3
 \end{aligned}$$

A finite series converges for all x , so $R = \infty$.

(b)

n	$f^{(n)}(x)$	$f^{(n)}(-3)$
0	$1/x$	$-1/3$
1	$-1/x^2$	$-1/3^2$
2	$2/x^3$	$-2/3^3$
3	$-6/x^4$	$-6/3^4$
4	$24/x^5$	$-24/3^5$
$\vdots$	$\vdots$	$\vdots$

$$\begin{aligned}
 f(x) &= \frac{1}{x} = \sum_{n=0}^{\infty} \frac{f^{(n)}(-3)}{n!} (x+3)^n \\
 &= \frac{-1/3}{0!} (x+3)^0 + \frac{-1/3^2}{1!} (x+3)^1 + \frac{-2/3^3}{2!} (x+3)^2 \\
 &\quad + \frac{-6/3^4}{3!} (x+3)^3 + \frac{-24/3^5}{4!} (x+3)^4 + \cdots \\
 &= \sum_{n=0}^{\infty} \frac{-n!/3^{n+1}}{n!} (x+3)^n = - \sum_{n=0}^{\infty} \frac{(x+3)^n}{3^{n+1}}
 \end{aligned}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x+3)^{n+1}}{3^{n+2}} \cdot \frac{3^{n+1}}{(x+3)^n} \right| = \lim_{n \rightarrow \infty} \frac{|x+3|}{3} = \frac{|x+3|}{3} < 1 \quad \text{for convergence,}$$

so $|x+3| < 3$ and $R = 3$.

(c)

n	$f^{(n)}(x)$	$f^{(n)}(\pi/2)$
0	$\sin x$	1
1	$\cos x$	0
2	$-\sin x$	-1
3	$-\cos x$	0
4	$\sin x$	1
$\vdots$	$\vdots$	$\vdots$

$$\begin{aligned}
 f(x) &= \sin x = \sum_{k=0}^{\infty} \frac{f^{(k)}(\pi/2)}{k!} \left(x - \frac{\pi}{2}\right)^k \\
 &= 1 - \frac{(x - \pi/2)^2}{2!} + \frac{(x - \pi/2)^4}{4!} - \frac{(x - \pi/2)^6}{6!} + \cdots \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{(x - \pi/2)^{2n}}{(2n)!}
 \end{aligned}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left[\frac{|x - \pi/2|^{2n+2}}{(2n+2)!} \cdot \frac{(2n)!}{|x - \pi/2|^{2n}} \right] = \lim_{n \rightarrow \infty} \frac{|x - \pi/2|^2}{(2n+2)(2n+1)} = 0 < 1 \quad \text{for all } x, \text{ so } R = \infty.$$

(d)

n	$f^{(n)}(x)$	$f^{(n)}(16)$
0	$\sqrt{x}$	4
1	$\frac{1}{2}x^{-1/2}$	$\frac{1}{2} \cdot \frac{1}{4}$
2	$-\frac{1}{4}x^{-3/2}$	$-\frac{1}{4} \cdot \frac{1}{4^3}$
3	$\frac{3}{8}x^{-5/2}$	$\frac{3}{8} \cdot \frac{1}{4^5}$
4	$-\frac{15}{16}x^{-7/2}$	$-\frac{15}{16} \cdot \frac{1}{4^7}$
$\vdots$	$\vdots$	$\vdots$

$$\begin{aligned}
 f(x) &= \sqrt{x} = \sum_{n=0}^{\infty} \frac{f^{(n)}(16)}{n!} (x-16)^n \\
 &= \frac{4}{0!} (x-16)^0 + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{1!} (x-16)^1 - \frac{1}{4} \cdot \frac{1}{4^3} \cdot \frac{1}{2!} (x-16)^2 \\
 &\quad + \frac{3}{8} \cdot \frac{1}{4^5} \cdot \frac{1}{3!} (x-16)^3 - \frac{15}{16} \cdot \frac{1}{4^7} \cdot \frac{1}{4!} (x-16)^4 + \dots \\
 &= 4 + \frac{1}{8}(x-16) + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3)}{2^n 4^{2n-1} n!} (x-16)^n \\
 &= 4 + \frac{1}{8}(x-16) + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3)}{2^{5n-2} n!} (x-16)^n
 \end{aligned}$$

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(-1)^n 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)(x-16)^{n+1}}{2^{5n+3}(n+1)!} \cdot \frac{2^{5n-2}n!}{(-1)^{n-1} 1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3)(x-16)^n} \right| \\
 &= \lim_{n \rightarrow \infty} \frac{(2n-1)|x-16|}{2^5(n+1)} = \frac{|x-16|}{32} \lim_{n \rightarrow \infty} \frac{2-1/n}{1+1/n} = \frac{|x-16|}{32} \cdot 2 \\
 &= \frac{|x-16|}{16} < 1 \quad \text{for convergence, so } |x-16| < 16 \text{ and } R = 16.
 \end{aligned}$$